

Based on the findings of five practitioners in the previous stage, refinements were made to the document-based taxonomy for urban planning policy discourse presented in Table I. This process resulted in the final taxonomy as a framework of aspects for the application of Aspect-Based Sentiment Analysis (ABSA), as presented in Table III. This taxonomy encompasses eight main categories: land use and land management; urban infrastructure and mobility; housing and basic services; environment, climate and disaster management; local economy and livelihoods; human development and social welfare; governance and public services; and social, cultural, and public order. Each category is accompanied by a definition, the scope of issues or sub-aspects, category boundaries or exclusion rules, and its use in advanced ABSA as an aspect label to identify, group, and analyze public opinion based on the focus of the development issues under discussion.

TABLE III. FINAL TAXONOMY AS AN ASPECT SCHEME FOR ABSA

Final Categories	Definition	Included Issues or Sub-aspects	Category Boundaries or Exclusion Rules	Use in Advanced ABSA
Land Use and Land Management	Opinions addressing planning, utilization, control, compliance, permitting, or conflicts related to land and spatial planning.	Regional Spatial Plans (RTRW), zoning, land use designations, land-use conversion, site permits, land use, strategic areas, buildings not in compliance with their designated use, land disputes, and land administration.	Complaints regarding roads, drainage, or housing conditions are not included in this category, unless the primary focus of the opinion is the appropriateness of land use, land designation, or land-related issues.	Serves as a framework for identifying and categorizing public opinion regarding spatial planning and land use.
Urban Infrastructure and Mobility	Opinions addressing the availability, quality, maintenance, safety, accessibility, and connectivity of physical infrastructure and urban transportation.	Roads, bridges, sidewalks, street lighting, irrigation, traffic congestion, public transportation, terminals, parking, traffic signs, road safety, and connectivity.	Opinions focusing on drinking water, sanitation, wastewater, stormwater drainage, or waste management are included in the Housing and Basic Services category.	Serves as a label to identify opinions related to physical infrastructure and urban mobility.
Housing and Basic Services	An opinion piece discussing housing suitability, settlement conditions, and the availability and quality of basic services in residential areas.	Housing suitability, substandard housing (RTLH), housing, slum areas, infrastructure, facilities, and public utilities; clean water, municipal water supply (PDAM), sanitation, septic tanks, wastewater, neighborhood drainage, waste, temporary waste collection sites (TPS), and permanent waste disposal sites (TPA).	Widespread pollution, green open spaces, conservation, and disaster risks are included in the Environment, Climate, and Disasters category if the opinion does not specifically focus on residential services.	This serves as a label to group opinions regarding housing quality, settlements, and basic urban services.
Environment, Climate, and Disasters Management	Opinion pieces discussing environmental protection and quality, pollution, green open spaces, climate impacts, disaster risks, preparedness, and emergency response.	Water and air quality, rivers, green open spaces, pollution, hazardous waste, flooding, coastal erosion, earthquakes, tsunamis, landslides, fires, mitigation, evacuation, and disaster preparedness.	Complaints regarding household waste, drainage systems, clean water, or sanitation are categorized under "Housing and Basic Services" if the opinion's primary focus is on the quality or availability of basic services.	Serves as a label to identify opinions regarding environmental quality, climate change, and disaster resilience.
Local Economy and Livelihoods	Opinion pieces discussing economic activity, employment opportunities, local businesses, trade, investment, food production, agriculture, fisheries, tourism, and affordability.	SMEs, cooperatives, employment, unemployment, markets, business capital, food prices, trade, investment, industry, agriculture, livestock, fisheries, fishermen, tourism, and the creative economy.	Social assistance, health care, education, or services for vulnerable groups are included in the Human Development and Social Welfare category if the opinion piece does not focus on economic activities or livelihoods.	This serves as a label to categorize opinions regarding the regional economy and people's livelihoods.
Human Development and Social Welfare	Opinion pieces discussing education, health, poverty, social assistance, social protection, and services for vulnerable groups.	Schools, teachers, scholarships, education, hospitals, community health centers, BPJS, stunting, poverty, social assistance, people with disabilities, the elderly, women, children, and social rehabilitation.	Complaints regarding ID cards (KTP), family cards (KK), certificates, permits, government applications, or administrative responses from government officials are	Serves as a label to identify opinions regarding the quality of human development, social

Final Categories	Definition	Included Issues or Sub-aspects	Category Boundaries or Exclusion Rules	Use in Advanced ABSA
			categorized under Governance and Public Services.	services, and community welfare.
Governance and Public Services	Opinion pieces discussing government administration, administrative services, licensing, transparency, accountability, digitization of services, government data, and the response of government officials.	The performance of regional government agencies (OPD), the Regional Development Planning Agency (Bappeda), public services, ID cards (KTP), family cards (KK), certificates, licensing, government applications, the Electronic Government System (SPBE), data, transparency, the budget, bureaucracy, and service complaints.	Crime, public order, social conflicts, enforcement of local regulations, and environmental safety are included in the Social, Cultural, and Public Order category.	These serve as labels to group opinions regarding the quality of governance and public services.
Social, Cultural, and Public Order	Opinions discussing security, public order, legal compliance, citizen participation, social conflict, local culture, religion, and community identity.	Crime, Satpol PP, local regulations, neighborhood watch posts, community safety, Musrenbang, residents' concerns, social conflicts, Tabot, customs, the arts, cultural heritage sites, local history, places of worship, and community harmony.	Opinions that primarily discuss population administration, licensing, digital services, or bureaucratic processes are categorized under Governance and Public Services.	These serve as tags to identify opinions regarding social conditions, culture, security, participation, and public order.

Note: In opinion pieces that address more than one development issue, the primary category is determined based on the most dominant core issue. Other issues may be noted as supporting context or labeled as secondary aspects. This rule is used to address cross-sectoral issues in accordance with input from practitioners.